CSE110 Spring'25 Final Ques&Solve

Computer Programing (Java-1) (BRAC University)

No. of Pages	3
No. of Questions	4
Total Marks	30
Time: 90 minutes	

Department of Computer Science and Engineering
FINAL EXAMINATION Spring 2025
CSE 110: Programming Language I

A

- ✓ Write theory teacher's Name/Initial on top of the answer script in LARGE FONT.
- ✓ Answer all questions. Use **back part** of the answer script for rough work.
- ✓ Answer Question **1 & 2** at the **beginning part** of the answer script.
- ✓ Figure in bracket [] next to each question indicates marks for that question.
- ✓ At the end of exam, put **question paper** inside answer script and **return both.**
- ✓ Understanding the question is part of the exam, **please do not ask questions.**

No washroom breaks

Section: ___ **ID:** __________ **Name in CAPITAL:** ____________________

Question 1 [CO2] [10 Points]
[Answer on the answer-script]

Write a Java program that simplifies a basic algebraic expression provided as a string. The expression will consist of:

- Whole numbers (both single and multi-digit, like 7, 13, 103, etc.)
- The addition symbol + and subtraction symbol -
- Exactly two types of variables: x and y
- The input expression will contain no spaces, no parentheses, and no multiplication symbols.
- Each term in the expression will be of the form [coefficient][variable], such as 3x, -4y, +5x, etc.

Your task is to take the expression as String input, parse the expression and combine like terms. That means, all x terms should be added/subtracted together, and the same for y terms. Finally, you should output the simplified form as a string in the format. **[You are not allowed to use Integer.parseInt() method]**

Sample Input	Output
-7x-13x+2y-11y-4x	-24x-9y
100x-13y-7x+22x-13y+113y	115x+87y

Question 2 [CO3] [10 Points]
[Answer on the answer-script]

Suppose, you are given an integer array named arr. Your task is to find an index i such that arr[i] equals sum of all elements to both the left and right of i. In case of multiple valid indices, print all. If there is none, print "Not Applicable". You can assume that the length of the array will always be greater than two and index 0 & index (arr.length - 1) will always be invalid. **Your program should work for any given integer array with length greater than two.**

Given Array	Sample Output	Explanation
int[] arr = {2, 2, 1, 0, 1, 6, 2, 4};	5	arr[5] = 6 = 2 + 2 + 1 + 0 + 1 = 2 + 4

`int[] arr = {7, -5, 3, 2, 1};`	Not Applicable	There is no such index possible where the mentioned criteria are fulfilled.
`int[] arr = {0, 0, 0, 0, 0, 0};`	1 2 3 4	Since index 0 and index 5 are invalid, all the remaining indices fulfill the mentioned criteria

Question 3 [CO4] [5 Points]

[Answer on the question paper]

The code below is designed to print a **left-justified hollow Triangle**. However, it contains **4 errors** and **2 missing parts**. Analyze the code, identify the errors, and correct them. Also, fill in the missing parts of the code. Provide your answers in the specified format.

```
1  public class A{
2   public static void main(String args){
3     int num = 6;
4     hollowTriangle(int num);
5   }
6   public static String hollowTriangle(int n){
7     for (int row = 1; row <= num; row++) {
8        for (int col = 1;________; col++) {
9          if (col == 1 || col == row || row == n) {
10           System.out.print(________);
11         } else {
12           System.out.print(" ");
13         }
14       }
15       System.out.println();
16     }
17   }
18 }
```

```
1
12
1 3
1  4
1   5
-123456-
```

Write your corrections in the table below. For better understanding, one error correction is shown.

Line Number	Fix
Line 2	public static void main(String args []){

Question 4 [CO4] [5 Points]
[Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. **Your answer will not be accepted without the workings. All of the outputs must be in the question paper.**

```java
public class methodTracing_A{
  public static void main(String [] args){
    m1(1);
  }
  public static int m1(int n){
    int b = 2;
    if (n<=5){
      b = m1(n+1) + m2(n+b);
      System.out.println(b);
    }else{
      return b;
    }
    return b;
  }
  public static int m2(int a){
    int b = a + a%2*3;
    return a-b*a;
  }
}
```

Output

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Section: ___ **ID:** _____________ **Name in CAPITAL:** ______________________________

Question 1 [CO2] [10 Points]
[Answer on the answer-script]

Write a Java program that simplifies a basic algebraic expression provided as a string. The expression will consist of:

- Whole numbers (both single and multi-digit, like 7, 13, 103, etc.)
- The addition symbol + and subtraction symbol -
- Exactly two types of variables: x and y
- The input expression will contain no spaces, no parentheses, and no multiplication symbols.
- Each term in the expression will be of the form [coefficient][variable], such as 3x, -4y, +5x, etc.

Your task is to take the expression as String input, parse the expression and combine like terms. That means, all x terms should be added/subtracted together, and the same for y terms. Finally, you should output the simplified form as a string in the format. **[You are not allowed to use Integer.parseInt() method]**

Sample Input	Output
-7x-13x+2y-11y-4x	-24x-9y
100x-13y-7x+22x-13y+113y	115x+87y

Question 2 [CO3] [10 Points]
[Answer on the answer-script]

Suppose, you are given an integer array named arr. Your task is to find an index i such that arr[i] equals sum of all elements to both the left and right of i. In case of multiple valid indices, print all. If there is none, print "Not Applicable". You can assume that the length of the array will always be greater than two and index 0 & index (arr.length - 1) will always be invalid. **Your program should work for any given integer array with length greater than two.**

Given Array	Sample Output	Explanation
int[] arr = {2, 2, 1, 0, 1, 6, 2, 4};	5	arr[5] = 6 = 2 + 2 + 1 + 0 + 1 = 2 + 4

int[] arr = {7, -5, 3, 2, 1};	Not Applicable	There is no such index possible where the mentioned criteria are fulfilled.
int[] arr = {0, 0, 0, 0, 0, 0};	1 2 3 4	Since index 0 and index 5 are invalid, all the remaining indices fulfill the mentioned criteria

Question 3 [CO4] [5 Points]

[Answer on the question paper]

The code below is designed to print a **hollow square**. However, it contains **4 errors and 2 missing parts**. Analyze the code, identify the errors, and correct them. Also, fill in the missing parts of the code. Provide your answers in the specified format.

```
12345
1   5
1   5
1   5
12345
```

```
1  public class B {
2   public static void main(String args)  {
3     int N = 5;
4     hollowSquare(N);
5   }
6   public static int hollowSquare(int n){
7     for (int r = 1; __________; r++) {
8       for (int c = 1; c <= N; c++) {
9         if (r == 1 || _________ || c == 1 || c == n) {
10          System.out.print(r);
11        } else {
12          System.out.print(" ");
13         }
14      }
15      System.out.println();
16    }
17  }
18 }
```

Write your corrections in the table below. For better understanding, one error correction is shown.

Line Number	Fix
	public static void main(String args []){

Question 4 [CO4] [5 Points]

[Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. **Your answer will not be accepted without the workings. All of the outputs must be in the question paper.**

```java
public class methodTracing_B{
  public static void main(String [] args){
    m1(1);
  }
  public static int m1(int n){
    int b = 2;
    if (n<=5){
      b = m1(n+1) + m2(n+b);
      System.out.println(b);
    }else{
      return b;
    }
    return b;
  }
  public static int m2(int a){
    int b = a%2*3 - a;
    return a-b*a;
  }
}
```

Output

CSE110 Spring 2025 Final Exam Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss it in the theory faculty slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.
You are free to use your best judgment while evaluating.

1.1 Set A and Set B Question 1 (Coding) Tentative Solution

```
import java.util.Scanner;
public class Q1 {
  public static void main(String[] args){
    Scanner sc = new Scanner(System.in);
    String inp = sc.nextLine();
    String sym = "+";
    int res_x = 0, res_y = 0;
    int s = 0, temp = 0;

    if(inp.charAt(0) == '-'){
      sym = "-";
      s++;
    }
    for(int i = s; i < inp.length(); i++){
      char c = inp.charAt(i);
      if(c >= '0' && c <= '9'){
        temp = temp * 10 + (c - '0');
      }else{
        if(sym.equals("+")){
          if(c == 'x'){
            res_x += temp;
          }else if(c == 'y'){
            res_y += temp;
          }
        }else{
          if(c == 'x'){
            res_x -= temp;
          }else if(c == 'y'){
            res_y -= temp;
          }
        }
        sym = c+"";
        temp = 0;
      }
    }
    if(res_y < 0){
      System.out.println(res_x + "x" + res_y + "y");
    }else{
      System.out.println(res_x + "x+" + res_y + "y");
    }
  }
}
```

CSE110 Spring 2025 Final Exam Tentative Solutions and Rubrics

1.2 Rubric for Set A and Set B Question 1 (Coding)

SL	Points to Meet	Marks (10)
1	Importing Scanner class, Creating Scanner object and Java syntaxes (i.e. class, static void main, semicolon, curly braces and parentheses)	0.5
2	Properly taking user input and initializing other variables	1
3	Handling the initial sign '+' or '-'	1
4	Iterating the loop correctly	1
5	Finding the digit and converting it to an integer number	1
6	Performing + for both x and y based on the *sym*	2
7	Performing - for both x and y based on the *sym*	2
8	Updating the *temp* and *sym* variables	0.5
9	Printing the output properly based on the conditions	1
	Total	**10**

CSE110 Spring 2025 Final Exam Tentative Solutions and Rubrics

2.1 Set A and Set B Question 2 (Coding) Tentative Solution

```
public class Q2{
  public static void main(String [] args){
    int[] arr = {2, 2, 1, 0, 1, 6, 2, 4};
    boolean found = false;
    for (int i = 1; i < arr.length - 1; i ++){ // excluding the first and last index
      int sum1 = 0;
      for (int j = 0; j < i; j++){
        sum1 += arr[j];
      }
      int sum2 = 0;
      for (int k = i + 1; k < arr.length; k++){
        sum2 += arr[k];
      }
      if (arr[i] == sum1 && arr[i] == sum2){
        found = true;
        System.out.println(i);
      }
    }
    if (!found){
      System.out.println("Not Applicable");
    }
  }
}
```

2.2 Rubric for Set A and Set B Question 2 (Coding)

SL	Points to Meet	Marks (10)
1	Java syntaxes (i.e. class, static void main, semicolon, curly braces, and parentheses)	0.5
2	Initializing variables *arr* and *found*	1
3	Properly iterating the loop, excluding the first and last index	1
4	Calculating the sum of all the left elements with a proper loop and updating sum1	2
5	Calculating the sum of all the right elements with a proper loop and updating sum2	2
6	Checking the equality of the left sum and the right sum	1
7	Updating *found* variable and printing the index/indices	1
8	Checking and printing the case for "Not Applicable"	1.5
	Total	10

CSE110 Spring 2025 Final Exam Tentative Solutions and Rubrics

3.1 Set A and Set B Question 3 (Error Correction) Tentative Solution

Set A

Line Number	Fix
Line 4	`hollowTriangle(num);`
Line 6	`public static void hollowTriangle(int n){`
Line 7	`for (int row = 1; row <= n; row++) {`
Line 8	`for (int col = 1; col <= row; col++) {`
Line 10	`System.out.print(col);`

Set B

Line Number	Fix
Line 6	`public static void hollowSquare(int n){`
Line 7	`for (int r = 1; r <= n; r++) {`
Line 8	`for (int c = 1; c <= n; c++) {`
Line 9	`if (r == 1 || r == n || c == 1 || c == n) {`
Line 10	`System.out.print(c);`

3.2 Set A and Set B Question 3 (Error Correction) Rubric [5 Marks]

1. Each corrected line yields 1 mark
2. The sequence of the corrected lines doesn't need to match

CSE110 Spring 2025 Final Exam Tentative Solutions and Rubrics

4.1 Set A and Set B Question 4 (Tracing) Tentative Solution

SET A

Serial	Output
1	-61
2	-91
3	-126
4	-138
5	-153

SET B

Serial	Output
1	37
2	79
3	94
4	114
5	117

4.2 Set A and Set B Question 4 (Tracing) Rubric [5 Marks]

3. Each output yields 1 mark
4. The sequence of outputs must match
5. If no calculation/trace table is found anywhere in the script or the question, the student should receive 0 (zero) out of 5